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Capture the Flag 2: Widgets

Goal: Turn the single-card application into a simple multi-section interface.

Note from the instructor before you start:

These activities involve using your own problem solving and researching skills. We are strictly to implement NOT TO USE AI to generate the code for you. You may only use AI for RESEARCHING the syntax or usage, but NEVER THE CODE ITSELF.

Consequence: Immediate 0 score in your grade for this activity.

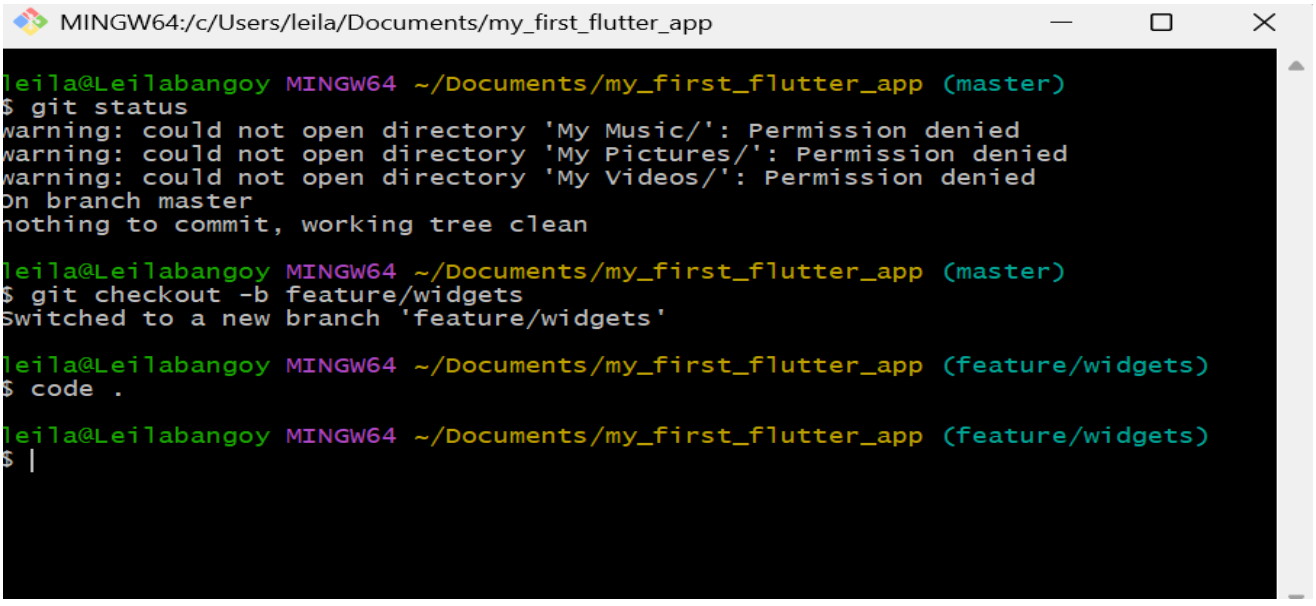
(I know what AI-generated code looks like, don't even think about it :>)

🏴 Flag 0 – Create Your Feature Branch

Objective: Create a feature branch before modifying your application. Name the branch feature/widgets

Screenshot:

- Terminal showing `feature/widgets`



🏴 Flag 1 – Break It Apart

Objective: Separate your profile information into individual `Text` widgets.

If your card currently looks like this:

```
Text(
  'Chiara Marie C. Canque\n'
  'BSIT - 3\n'
  'My First Flutter Application\n'
  'August 11, 2026',
)
```

Your task: Separate each piece of information into its own `Text` widget.

You should have something conceptually like:

Text → Name

Text → Course & Section

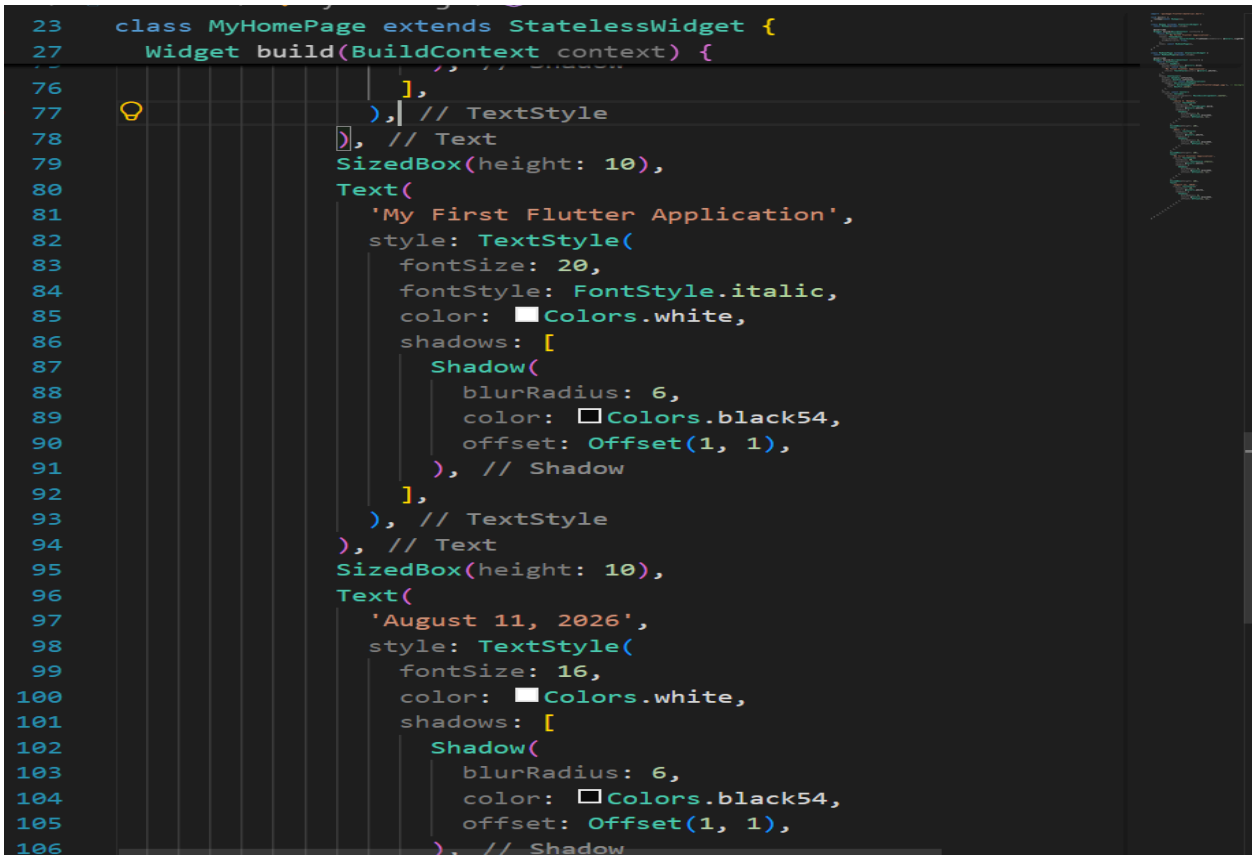
Text → Application Title

Text → Date

Do not worry about making it pretty yet.

Screenshot:

- Your code showing the separate `Text` widgets
- Your application



```
class MyHomePage extends StatelessWidget {
  Widget build(BuildContext context) {
                ],
              ), // TextStyle
            ), // Text
            SizedBox(height: 10),
            Text(
              'My First Flutter Application',
              style: TextStyle(
                fontSize: 20,
                fontStyle: FontStyle.italic,
                color: Colors.white,
                shadows: [
                  Shadow(
                    blurRadius: 6,
                    color: Colors.black54,
                    offset: Offset(1, 1),
                  ), // Shadow
                ],
              ), // TextStyle
            ), // Text
            SizedBox(height: 10),
            Text(
              'August 11, 2026',
              style: TextStyle(
                fontSize: 16,
                color: Colors.white,
                shadows: [
                  Shadow(
                    blurRadius: 6,
                    color: Colors.black54,
                    offset: Offset(1, 1),
```

🏴 Flag 2 – Stack the Information

Objective: Arrange your individual `Text` widgets vertically.

Research how to use:

`Column()`

Place your information inside the `Column`.

Your card should now have:

Name

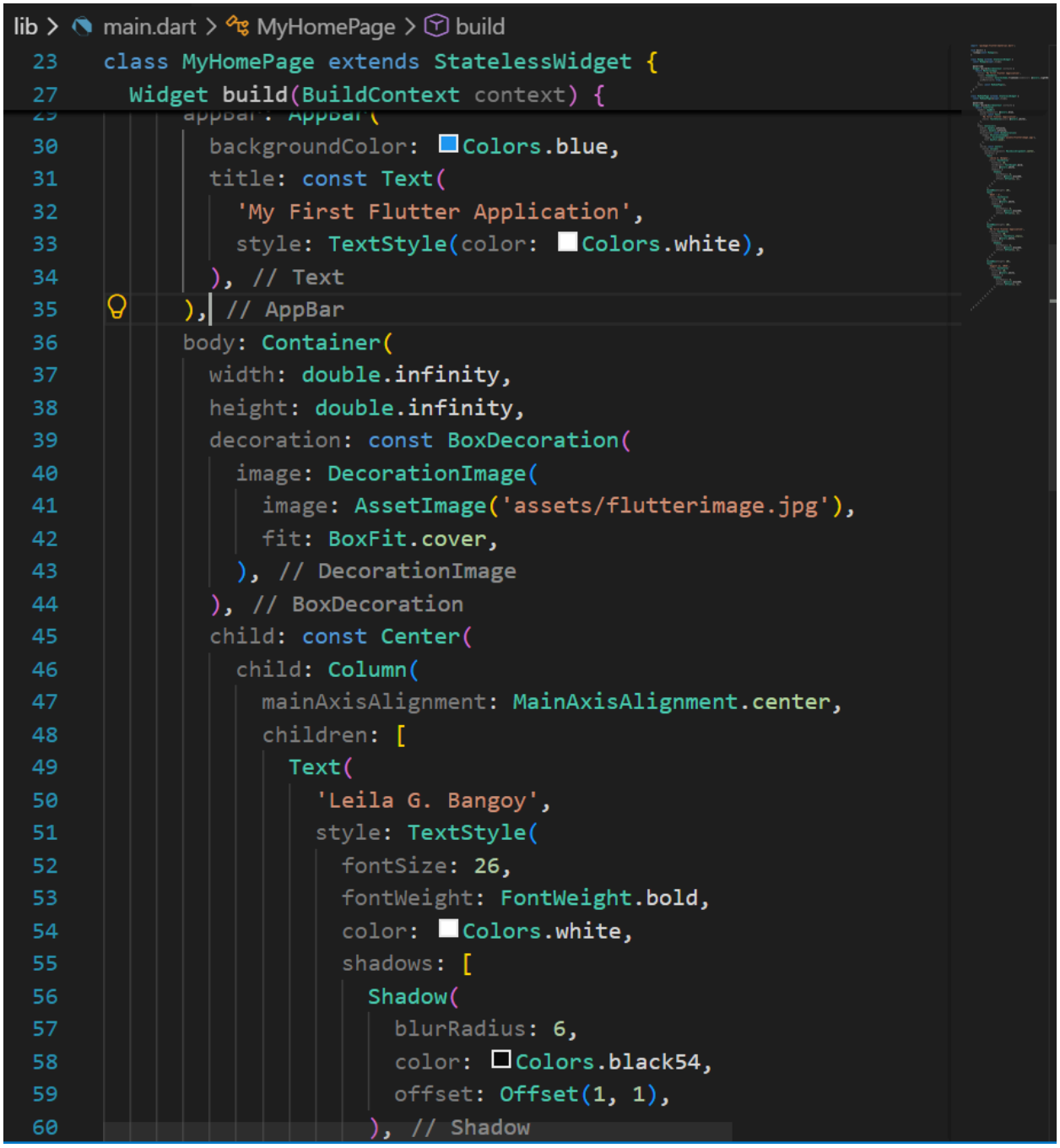
Course & Section

Application Title

Date

Screenshot:

- The `Column` in your code
- The four separate pieces of information displayed vertically



🏴 Flag 3 – Add Another Element

Objective: Add another individual widget to your `Column`.

Add something that belongs to your profile.

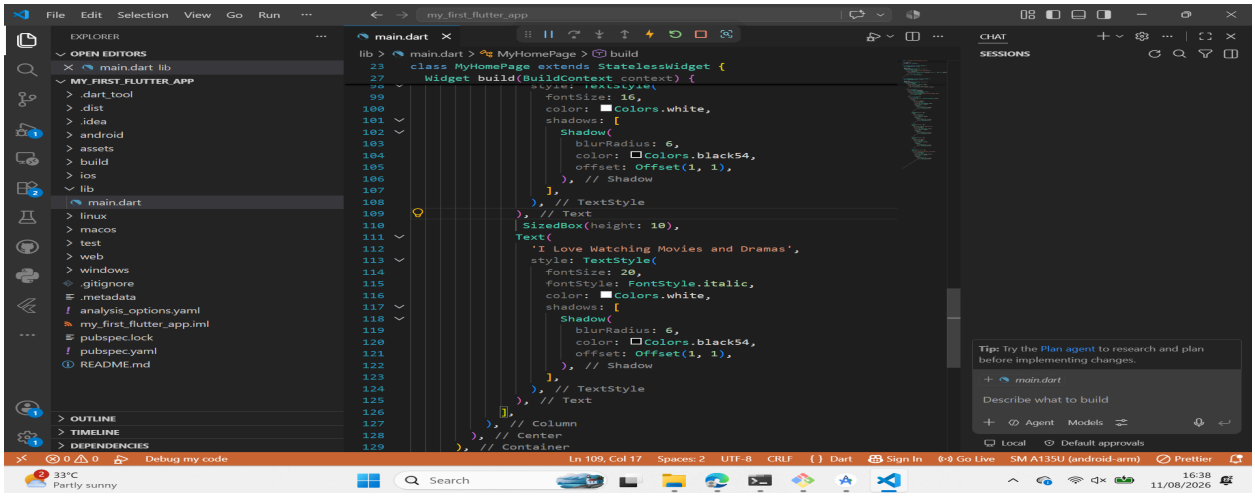
Examples:

Hobby

It must be its **own widget**, not additional text inside another `Text`.

Screenshot:

- New widget in your `Column`
- Updated application



🏴 Flag 4 – Put Two Things Side-by-Side

Objective: Discover the `Row` widget.

Create a new section containing **two pieces of information beside each other**.

For example:

Age: 21 Birthdate: August 11, 2003

Research:

`Row()`

Place the `Row` inside your existing `Column`.

Your structure should now resemble:

Column

├── Text

├── Text

├── Text

├── Text

├── Text

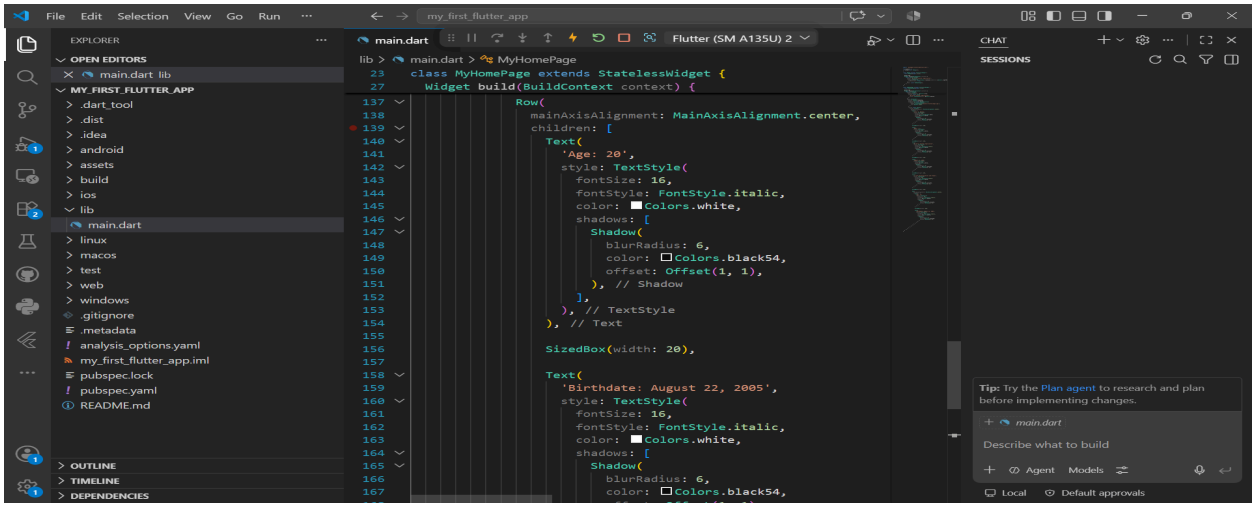
└── Row

    ├── Text

    └── Text

Screenshot:

- Your widget structure/code
- The two items displayed side-by-side



🏴 Flag 5 – Give It Some Space

Objective: Improve the spacing between your widgets.

Research:

`SizedBox()`

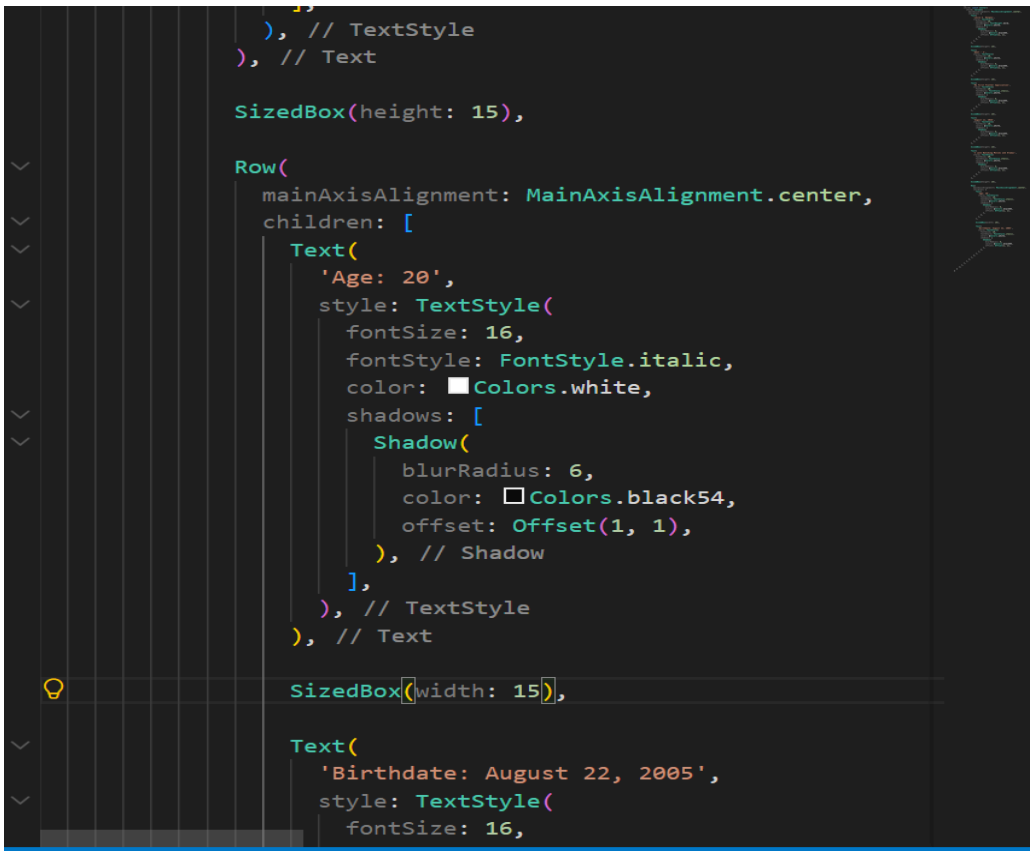
Add appropriate spacing between your:

- Image
- Name
- Course
- Other information
- Row

Challenge: Make the card easier to read **without changing its colors or fonts yet**.

Screenshot:

- `SizedBox` usage
- Updated application



🏴 Flag 6 – Make It Look Better

Objective: Apply styling to your individual widgets.

Now that your information is properly separated and positioned, improve the visual hierarchy.

Research how to modify:

`TextStyle()`

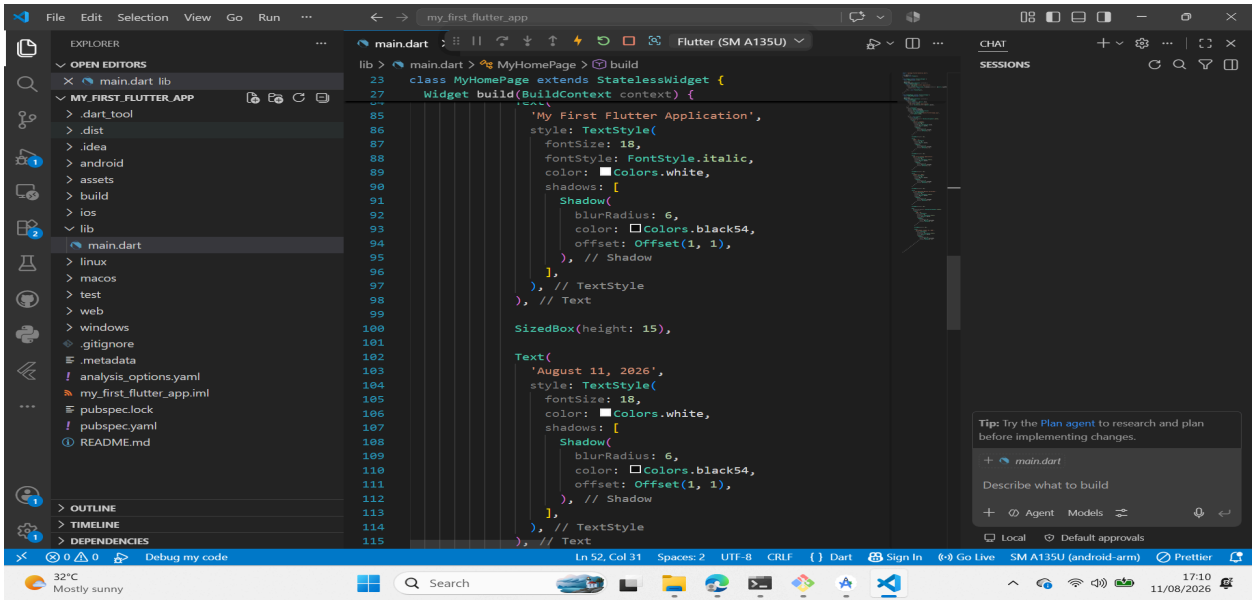
Apply different styling to at least **three** elements.

For example:

- Name → larger and bold
- Course → medium size
- Date → smaller
- Application title → italic

Screenshot:

- Your `TextStyle` changes
- Updated application



🏴 Flag 7 – Improve the Layout

Objective: Control the alignment of your widgets.

Research the properties of `Column` and `Row`.

Experiment with:

`mainAxisAlignment`

`crossAxisAlignment`

Change the alignment of your content so the card looks intentional rather than simply stacked.

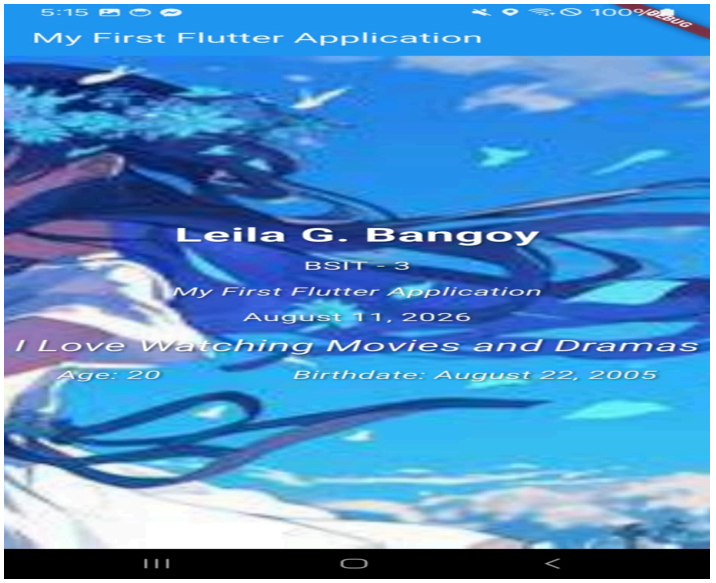
Screenshot:

- The properties you changed
- Updated application

```
                    offset: Offset(1, 1),
                  ), // Shadow
                ],
              ), // TextStyle
            ), // Text

            SizedBox(height: 15),

            Row(
              mainAxisAlignment: MainAxisAlignment.spaceEvenly,
              children: [
                Text(
                  'Age: 20',
                  style: TextStyle(
                    fontSize: 18,
                    fontStyle: FontStyle.italic,
                    color: Colors.white,
                    shadows: [
                      Shadow(
```



🏴 Flag 8 – Create a Second Card

Objective: Apply what you learned to create another section.

Create a second `Card` containing something different from your profile.

For example:

My Favorites

Favorite Game

Favorite Food

Favorite Movie

The second card should also use:

- Individual `Text` widgets
- A `Column`
- At least one `Row`
- Spacing

Screenshot:

- Code for the second card
- Both cards displayed

```
                    ],
                  ), // TextStyle
                ), // Text
              ],
            ), // Row
          ],
        ), // Column
      ), // Padding
    ), // Card

    SizedBox(height: 15),

    // ========================
    // SECOND CARD - FAVORITES
    // ========================

    Card(
      color: Colors.black54,
      child: Padding(
        padding: EdgeInsets.all(15),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [

            Text(
              'My Favorites',
              style: TextStyle(
                fontSize: 25,
                fontWeight: FontWeight.bold,
                color: Colors.white,
```

🏴 Flag 9 – Stack the Cards

Objective: Arrange your two cards vertically.

You now have:

Profile Card

↓

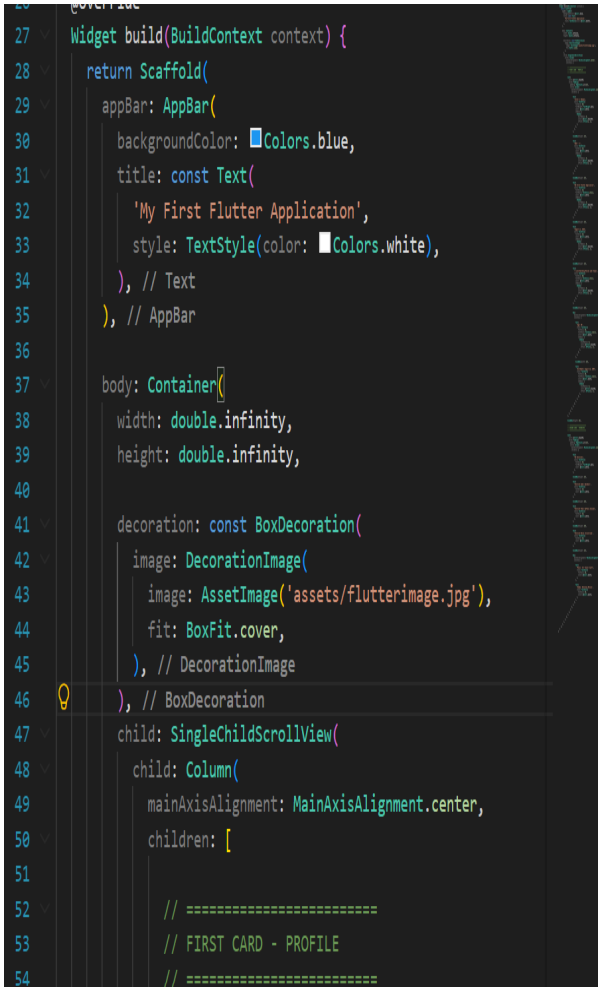
Favorites Card

Research how to place **multiple widgets inside the Flutter body**.

Use a layout widget to arrange your two cards vertically.

Screenshot:

- The widget containing both Cards
- Both cards visible in the application

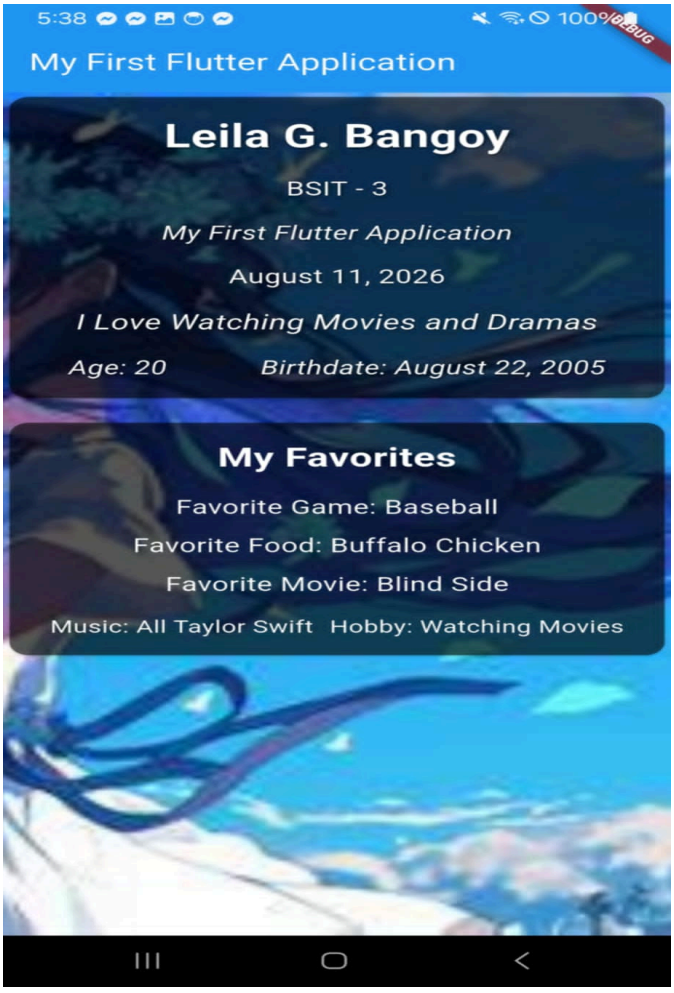


```
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      backgroundColor: Colors.blue,
      title: const Text(
        'My First Flutter Application',
        style: TextStyle(color: Colors.white),
      ), // Text
    ), // AppBar

    body: Container(
      width: double.infinity,
      height: double.infinity,

      decoration: const BoxDecoration(
        image: DecorationImage(
          image: AssetImage('assets/flutterimage.jpg'),
          fit: BoxFit.cover,
        ), // DecorationImage
      ), // BoxDecoration
      child: SingleChildScrollView(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [

            // ======================
            // FIRST CARD - PROFILE
            // ======================
```



🏴 Flag 10 – Widget Discovery

Objective: Discover a Flutter widget independently.

Research the Flutter widget catalog and find **one widget you haven't used yet**.

Examples:

- `Icon`
- `Divider`
- `CircleAvatar`
- `Chip`
- `ElevatedButton`

Add it somewhere appropriate in your application.

Screenshot:

- Your newly discovered widget in the code
- The widget visible in the application

```
Card(
  color: Colors.black54,
  child: Padding(
    padding: EdgeInsets.all(15),
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [

        Icon(
          Icons.favorite,
          color: const Color.fromARGB(255, 25, 102, 255),
          size: 200,
        ), // Icon

        SizedBox(height: 10),

        Text(
```

🏴 Flag 11 – Real Phone Test 📱

Objective: Run your completed application on a **physical** Android phone.

Research how to run Flutter on your phone.

Screenshot:

- Your completed application running on the physical phone (screenshot from your actual phone)

🏴 Flag 12 – Save Your Progress

Objective: Commit your completed Widget branch.

Follow proper standards of commit messages

Screenshot:

- Successful commit
- `feature/widgets` branch